

SMART INDIA HACKATHON 2026

THEME: CYBERSECURITY / AI • CATEGORY: SOFTWARE

ARGUS

DEFENSE CORE



A Hierarchical AI-Orchestrated Multi-Agent Cyber Defense Framework for Intelligent, Explainable and Autonomous Threat Detection and Response

DETECT



REASON



RESPOND



RECOVER



PROBLEM STATEMENT ID

[YOUR PS ID]

TEAM ID

[YOUR TEAM ID]

TEAM NAME

ARGUS AI

Problem Statement Title: ARGUS Defense Core — A Hierarchical AI-Orchestrated Multi-Agent Cyber Defense Framework for Intelligent, Explainable and Autonomous Threat Detection and Response



Understanding the Problem & Our Solution

ARGUS DEFENSE CORE



THE PROBLEM



Fragmented Security Tools

Independent tools generate alerts without coordinated reasoning.



Alert Overload

Large alert volumes make it hard to spot genuine attacks.



Slow Incident Response

Manual investigation and correlation delays response.



Limited Cross-Domain Visibility

Network, endpoint, malware & device events analyzed in isolation.



Reactive Defense

Systems detect and alert humans instead of reasoning & responding.

OUR SOLUTION — ARGUS DEFENSE CORE

Hierarchical AI-orchestrated multi-agent framework that coordinates specialized security agents to detect, investigate, explain and respond to cyber threats.



Real-Time Threat Detection

Continuously analyzes network, endpoint & device telemetry.



Multi-Agent Investigation

Specialized agents investigate different incident dimensions.



Hierarchical AI Orchestration

ARGUS Core assigns tasks & combines agent findings.



Explainable Threat Intelligence

Converts raw events into evidence-backed explanations.



Autonomous Defensive Planning

Generates response plans by severity & confidence.



Coordinated Response

Coordinates containment, isolation & recovery actions.

ARGUS USP STRIP



Multi-Agent
Cyber Defense



Explainable
Autonomous Response



Cross-Domain
Threat Correlation

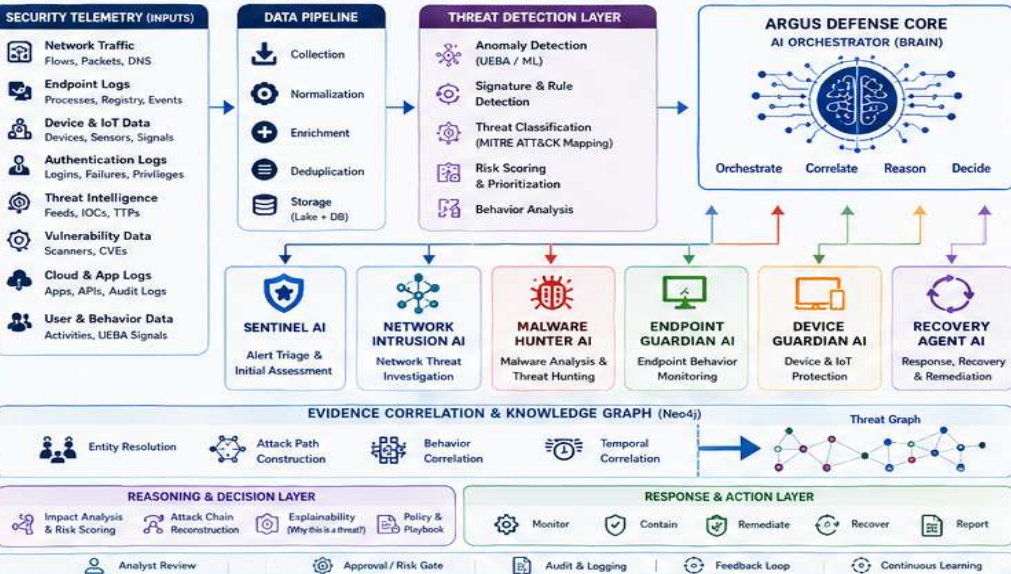


Detect → Reason →
Respond → Recover

1

ARGUS DEFENSE CORE – COMPLETE ARCHITECTURE

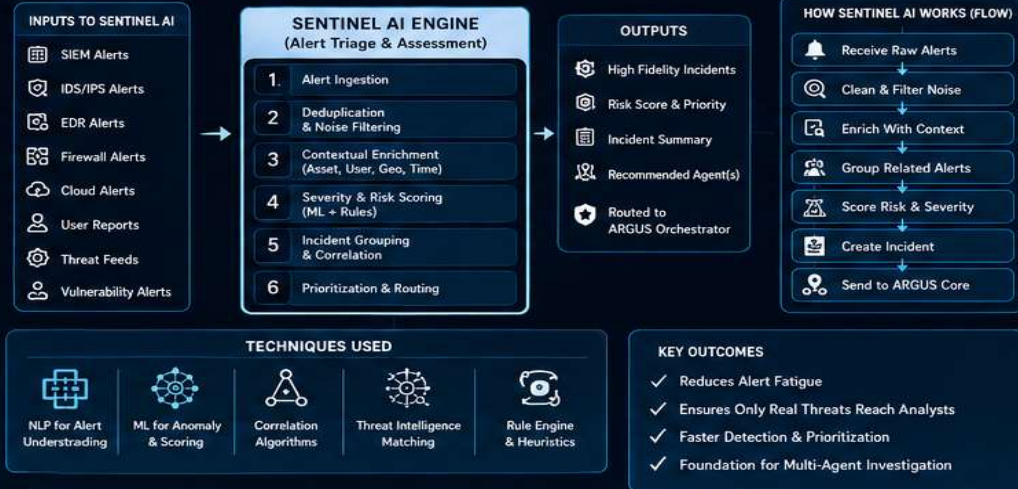
AI-ORCHESTRATED MULTI-AGENT CYBER DEFENSE FRAMEWORK



2

SENTINEL AI – ARCHITECTURE & WORKING

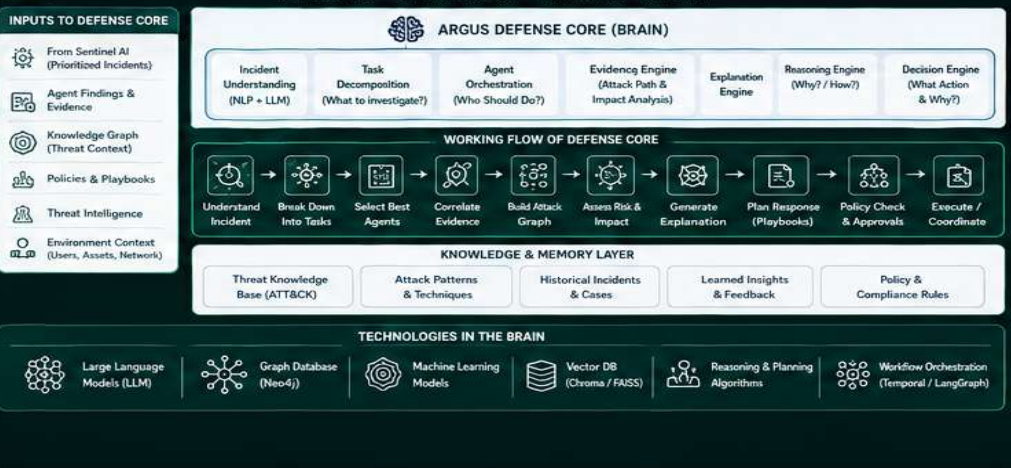
THE FIRST LAYER OF INTELLIGENCE – ALERT TRIAGE & PRIORITIZATION



3

ARGUS DEFENSE CORE (THE BRAIN) – ARCHITECTURE & WORKING

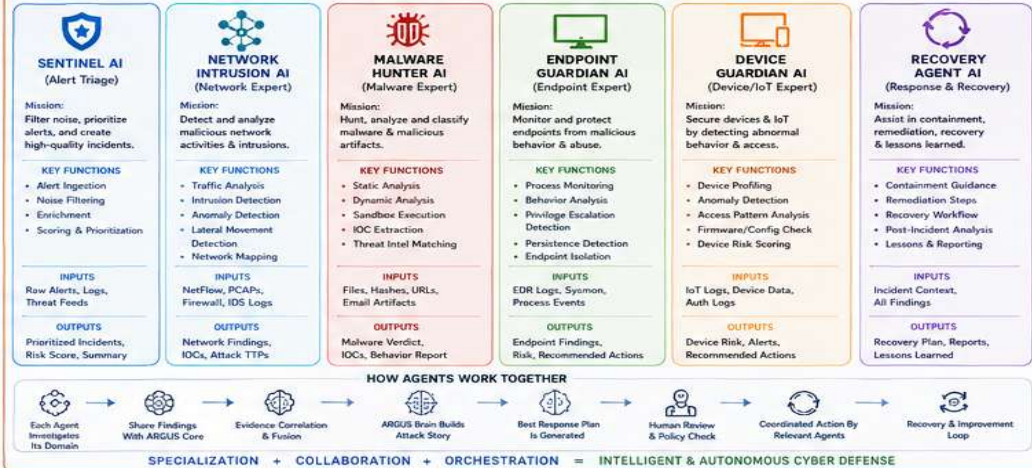
INTELLIGENCE • ORCHESTRATION • REASONING • DECISION



4

ARGUS AGENTS – DISTINCTION & WORKING

SPECIALIZED EXPERTS WORKING TOGETHER UNDER ARGUS CORE





Methodology & Technical Approach

HOW ARGUS WORKS



METHODOLOGY



1. Data Acquisition

Network traffic, endpoint & auth logs, device activity, malware indicators, threat intel, vulnerability data.



2. Data Preprocessing

Normalized, cleaned, timestamp-synchronized, deduplicated into machine-readable events.



3. Threat Detection

AI/ML identifies anomalies, intrusions, suspicious processes & malware (CICIDS2017, UNSW-NB15, NSL-KDD, MITRE ATT&CK).



4. ARGUS Orchestration Layer

Receives alert → understands context → decomposes investigation → assigns agents → correlates results → generates response plan.

SPECIALIZED AGENTS



Sentinel Agent

Initial detection, alert triage & event analysis.



Network Intrusion Agent

Analyzes traffic, connections & intrusion patterns.



Malware Hunter

Investigates files, processes, hashes & indicators.



Endpoint Guardian

Analyzes endpoint activity & host behavior.



Device Guardian

Monitors device-level anomalies.



Recovery Agent

Assists containment, remediation & recovery.

IMPLEMENTATION FLOW



Security Telemetry



Preprocessing



Threat Detection



ARGUS Core Orchestrator



Specialized Agents



Evidence Correlation



Explainability Layer



Response Planning



Impact & Benefits

WHY ARGUS MATTERS



IMPACT ON SECURITY TEAMS

- **Early Threat Detection**
Identifies suspicious activity before escalation.
- **Reduced Analyst Workload**
Automates repetitive investigation & correlation.
- **Faster Incident Response**
Shrinks time between detection and containment.
- **Explainable Decisions**
Evidence-backed explanations, not black-box alerts.
- **Continuous Monitoring**
Ongoing analysis across multiple attack surfaces.



SECURITY & OPERATIONAL BENEFITS

- **Risk Reduction**
Early detection & coordinated response cut impact.
- **Autonomous Investigation**
Agents investigate incident aspects simultaneously.
- **Cross-Domain Visibility**
Correlates network, endpoint, malware & device intel.
- **Faster SOC Operations**
Automates repetitive triage & investigation.
- **Incident Recovery**
Recovery Agent aids containment → remediation.



ECONOMIC & STRATEGIC GAINS

- **Reduced SOC Workload**
Automates repetitive alert investigation.
- **Lower Response Cost**
Earlier containment reduces prolonged-attack damage.
- **Scalable Architecture**
New agents added without system redesign.
- **Enterprise Applicability**
Enterprises, gov't, finance, healthcare, cloud.
- **Strategic Resilience**
Reactive security → proactive, AI-assisted defense.

EVALUATION METRICS (NOT ASSUMED FIGURES)



Precision / Recall
F1 / ROC-AUC



Mean Time to
Detect (MTTD)



Mean Time to
Respond (MTTR)



Agent Task
Success Rate



Safe-Action &
Human-Approval Rate



Feasibility & Viability

CAN IT ACTUALLY WORK?



ANALYSIS OF FEASIBILITY

- ✓ Proven AI Foundation — ML & anomaly detection widely used in cybersecurity
- 🗄️ Available Training Data — CICIDS2017, UNSW-NB15, NSL-KDD, MITRE ATT&CK
- 👥 Modular Architecture — agents independently developed, tested, deployed
- 🔌 Scalable Backend — FastAPI + PostgreSQL / Neo4j for events & relationships
- 🧠 LLM Integration — reasoning & summarization; deterministic controls stay critical-safe

KEY TECHNICAL CHALLENGES

False Positives

Multi-agent evidence correlation + confidence thresholds.

Large Data Volume

Event filtering, prioritization, async processing.

LLM Hallucination

Ground responses in retrieved evidence & structured data.

Autonomous Actions

Policy-based authorization + human-in-the-loop controls.

WHAT IF...?

01

AI makes a wrong decision?

Confidence thresholds, evidence verification, rule-based checks, human approval, audit logs.

02

Agent gives incorrect information?

ARGUS Core cross-validates via multiple agents, telemetry & confidence scoring.

03

LLM is manipulated?

Separate reasoning from execution; tool permissions, sanitization, structured outputs.

04

Attack evolves?

ARGUS dynamically launches further investigation as new evidence arrives.

05

An agent fails?

Core retries, reassigns agent, continues with partial evidence, or escalates.



Comparison & References

WHY ARGUS IS DIFFERENT

Feature	ARGUS Defense Core	Conventional Security
Architecture	Hierarchical multi-agent AI	Independent security tools
Detection	AI + multi-source correlation	Rule / signature / anomaly-based
Investigation	Multiple specialized agents	Primarily analyst-driven
Threat Correlation	Cross-agent evidence correlation	Often fragmented
Explainability	Evidence + reasoning + confidence	Alert-centric
Response	Coordinated response planning	Manual / tool-specific
Adaptability	Dynamic agent orchestration	Mostly predefined workflows
Recovery	Dedicated Recovery Agent	Separate manual process



RESEARCH & DATASET REFERENCES



MITRE ATT&CK

Adversary TTPs for threat modeling & attack mapping.



CICIDS2017

Network intrusion dataset: benign & malicious traffic.



UNSW-NB15

Network security dataset, modern attack categories.



NSL-KDD

Benchmark dataset for intrusion detection research.



VirusTotal

Threat intel / malware-analysis ecosystem.



MITRE D3FEND

Knowledge base of defensive techniques.



Neo4j

Graph DB for Users → Devices → Processes → Attacks.